

Overview of Chapter 2

2	What are applications of parallel and distributed systems?
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2.1	... from a Communication Perspective
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2.2	... from a Processing Perspective
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2.3	... from a Data Perspective
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2.4	... from a Privacy Perspective
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2.5	Mini-Seminar Topics
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Applications from a Data Perspective

- What are relevant applications from a data perspective?
 - Well ... basically everything!
- What are typical data characteristics?
 - There are no characteristic data. Anything goes.
 - From structured data (e.g., relational tables) via semi-structured data (e.g., documents) to unstructured data (e.g., multimedia) – and any combination in complex data models
 - From static data (e.g., immutable data) to “fast” data (e.g., continuous data streams)
 - From homogeneous data (all have the same structure) to heterogeneous data (all different)

Big Data in Today's Business and Private Use

- By 2025, there will more than 180 Zettabytes ($180 \cdot 10^{21}$) of data in the digital universe – out of which only ~10% is “original” data
- 90% of all data has been created in the last two years
- Today it would take a person approximately 181 million years to download all the data from the Internet
- In 2024, approximately 402.74 million terabytes of data were generated daily
- In 2020, every person generated ~1.7 megabytes in just a second
- In 2012, only 0.5% of all data was analyzed
- Internet users generate about 2.5 quintillion bytes (= 0.0025 Zettabytes) of data each day
- In 2019, Internet users spent 1.2 billion years online
- 97.2% of organizations are investing in big data and AI

Sources: <https://rivery.io/blog/big-data-statistics-how-much-data-is-there-in-the-world/> and [statista.com](https://www.statista.com)

Applications from a Data Perspective ...

- For structured data perspective, there are mainly two types of applications
- Serving the daily business: **OLTP (Online Transaction Processing)**
 - Transactional access to data, typically for creating (also updating) and subsequently reading data
 - Short read/write accesses to few data records
 - Example of OLTP systems: accounting systems, inventory control, stock trading, flight booking, etc.
 - Requires: **optimization** (performance), **consistency guarantees**, **fault tolerance**, etc.

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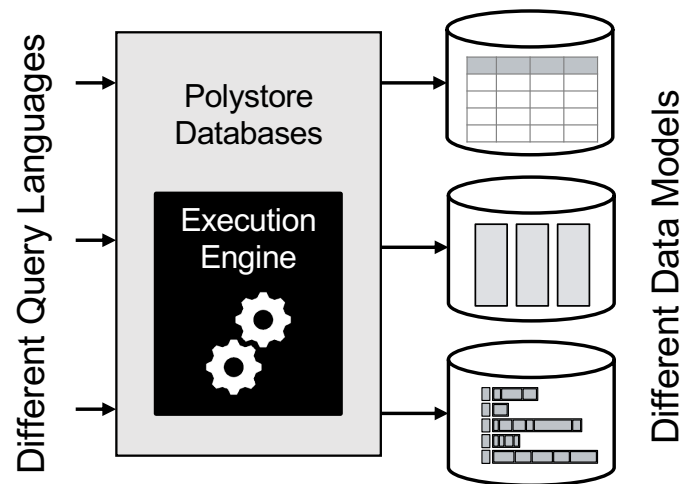
... Applications from a Data Perspective

- For structured data perspective, there are mainly two types of applications
- Doing analytics: **OLAP (On-line Analytical Processing)**
 - Consolidating, viewing, analyzing data in several dimensions, creation of reports (decision support)
 - Long and complex read transactions on many data records
 - Example: „Which goods can we recommend to a particular customer?“
 - Requires: **data integration**, complex queries, **ML**, etc.

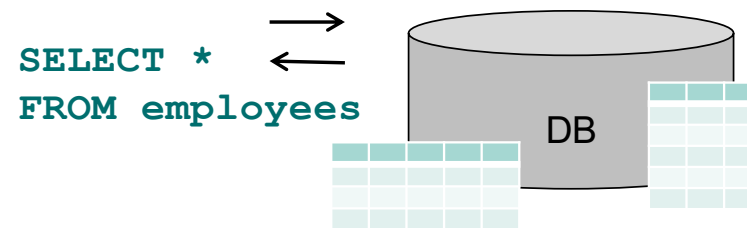


Mixed Workloads, Multiple Data Models – all in ONE

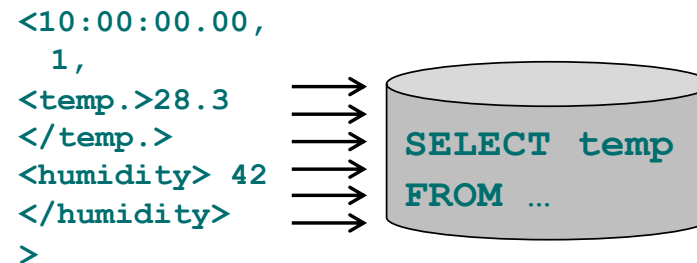
- Multi-Model Data Management
 - Mixed OLAP & OLTP Workloads
 - Different Query Languages
 - Boolean Queries & Similarity Search



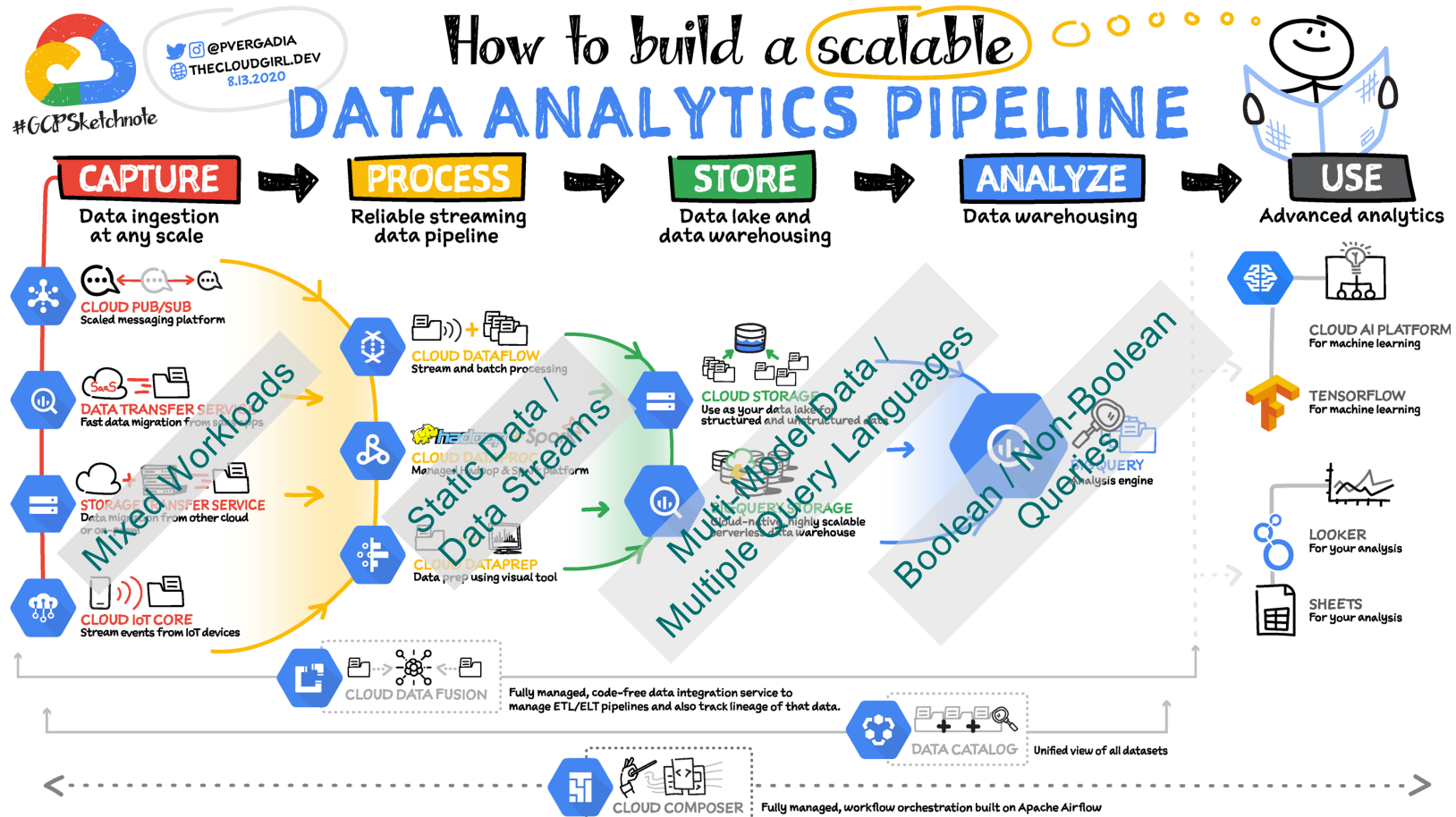
- Different Data Characteristics
 - Traditional: persistent data, transient queries



- Streams: transient data, persistent queries



Example: Data Science / Data Analytics



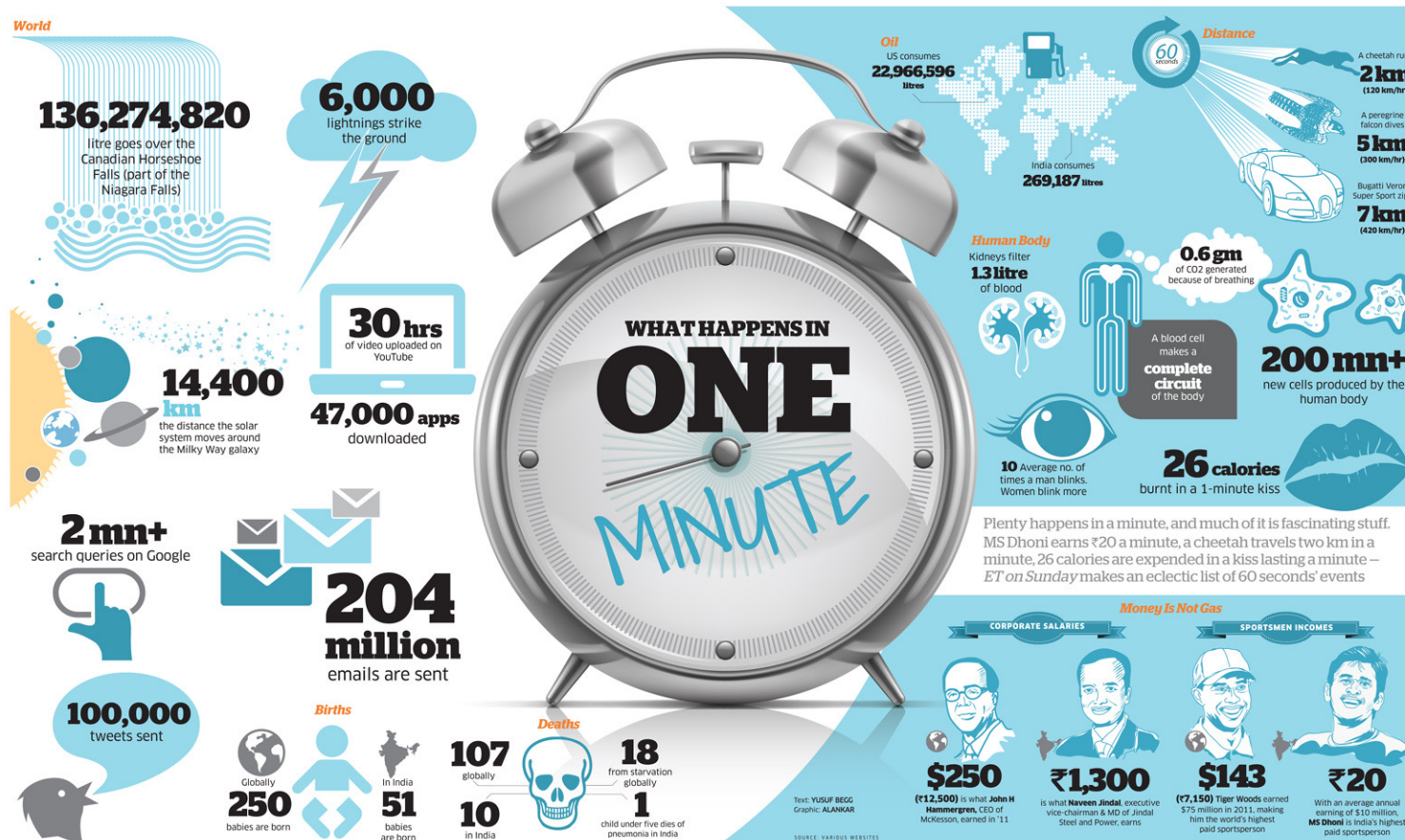
Source: <https://www.freecodecamp.org>

Big Data and Social Media

- Google processes 8.5 billion searches search queries per day (or 0.1 million per second).
- About 1'000 computers are involved in answering every Google query
- Each minute, 300 new hours of video show up on YouTube, leading to more than 1 billion gigabytes (1 exabyte) of data
- People share more than 100 terabytes of data on Facebook daily. Every minute, users send 31 million messages and view 2.7 million videos.
- WhatsApp users exchange up to 65 billion messages daily.
- Smart devices (for example, fitness trackers, sensors, Amazon Echo) produce 5 quintillion bytes of data daily. In 5 years, we can expect for the number of these gadgets to be more than 50 billion.
- More than 30% of these data will be uploaded to the cloud

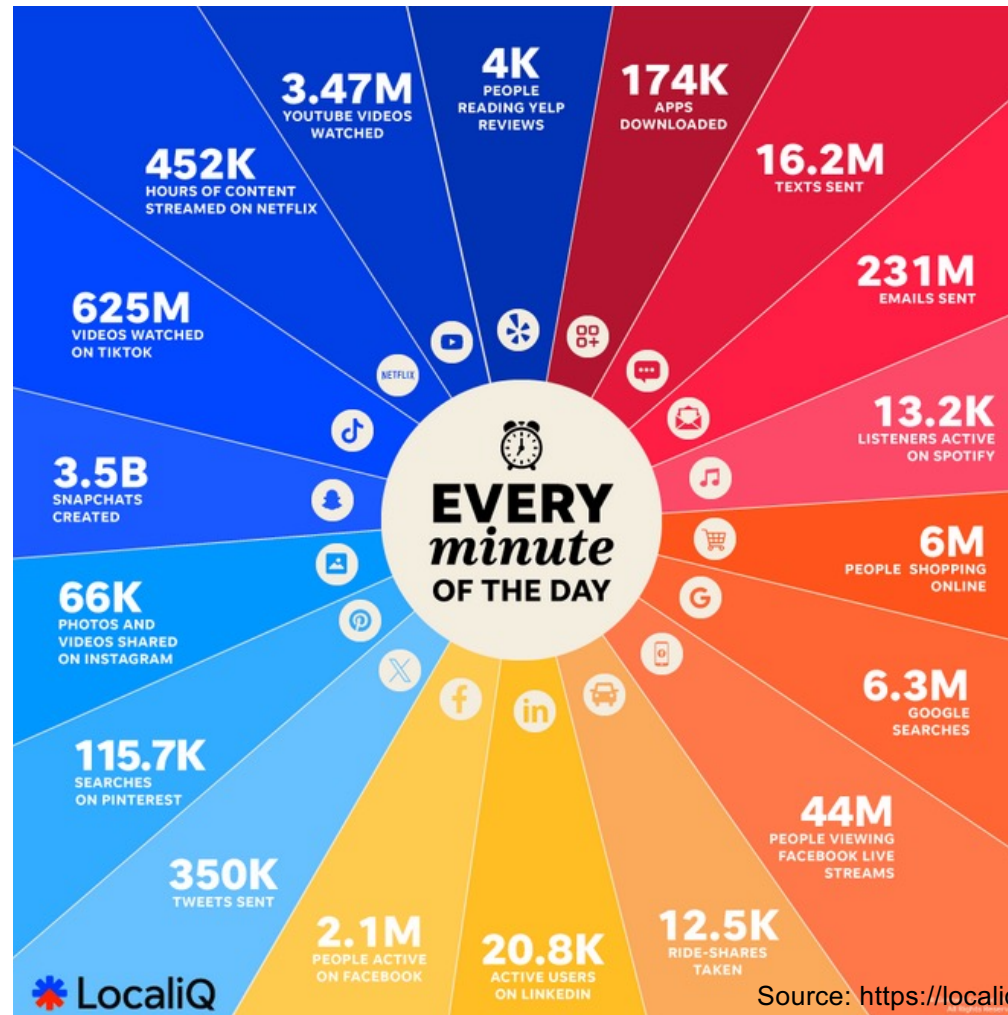
Sources: <https://hostingtribunal.com/blog/big-data-stats/> and <https://earthweb.com/how-many-google-searches-per-second/>

What Happens in one Minute?



Source: The Economic Times

The Internet in 2025 in Every Minute



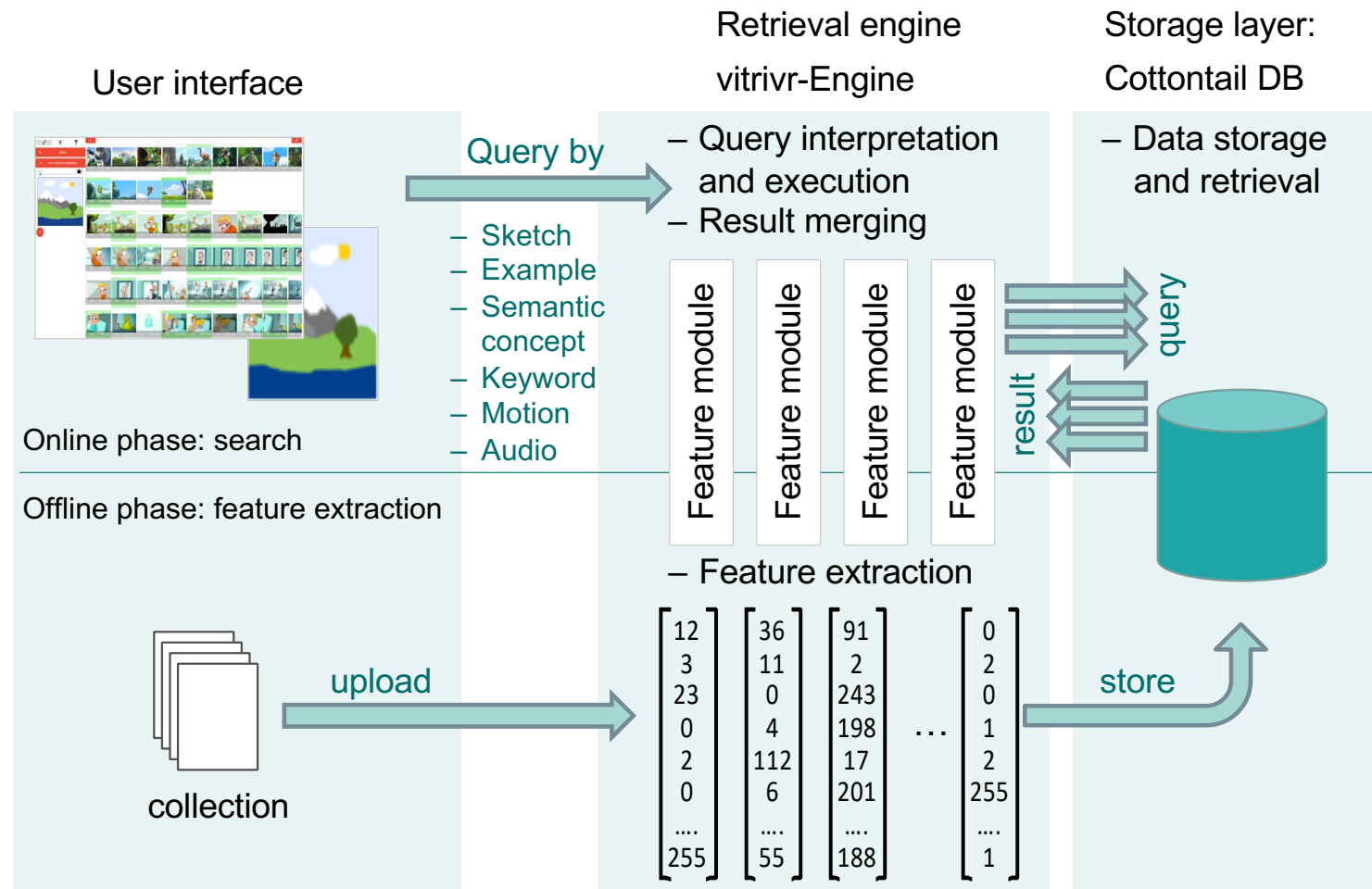
Source: <https://localiq.com/blog/what-happens-in-an-internet-minute/>

Big Data and Search Engines

- In 1999, it took Google one month to crawl and build an index of about 50 million pages. In 2012, this was done in less than 1 minute.
- 16% to 20% of queries that get asked every day have never been asked before.
- A single Google query uses 1'000 computers in 0.2 seconds to retrieve an answer.

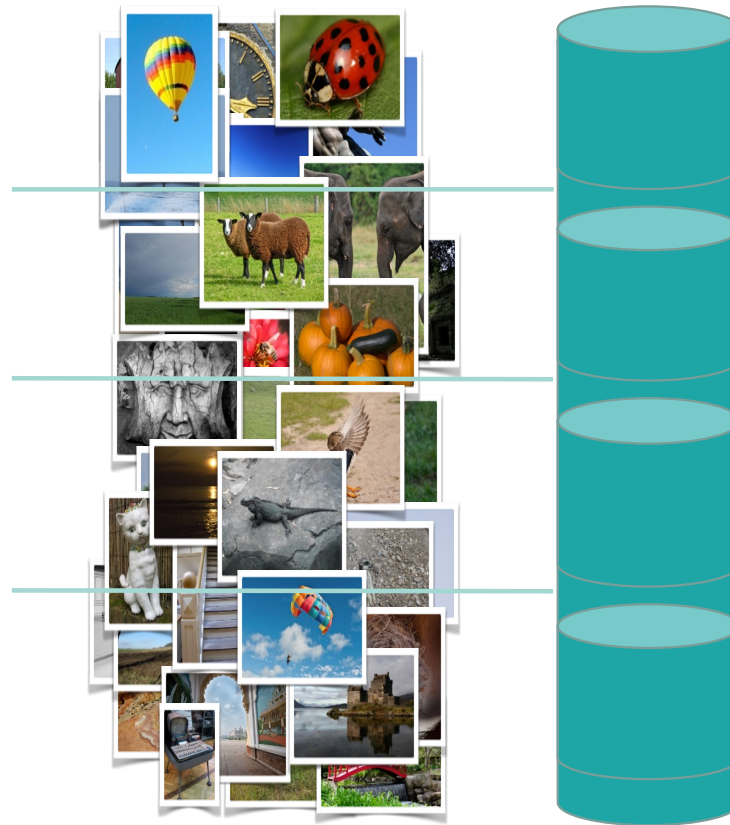
Source: <http://www.internetlivestats.com/google-search-statistics/>

Retrieval Applications: Example vitrivr



Retrieval Applications: Collection sizes

- Example: YFCC100M
 - collection of 99.2 million Creative Commons photos and 0.8 million CC-videos together with metadata (manually added tags, etc.)
 - total size is about 17 TB
 - partitioning and distribution, parallelization via Map/Reduce
 - several specialized index structures

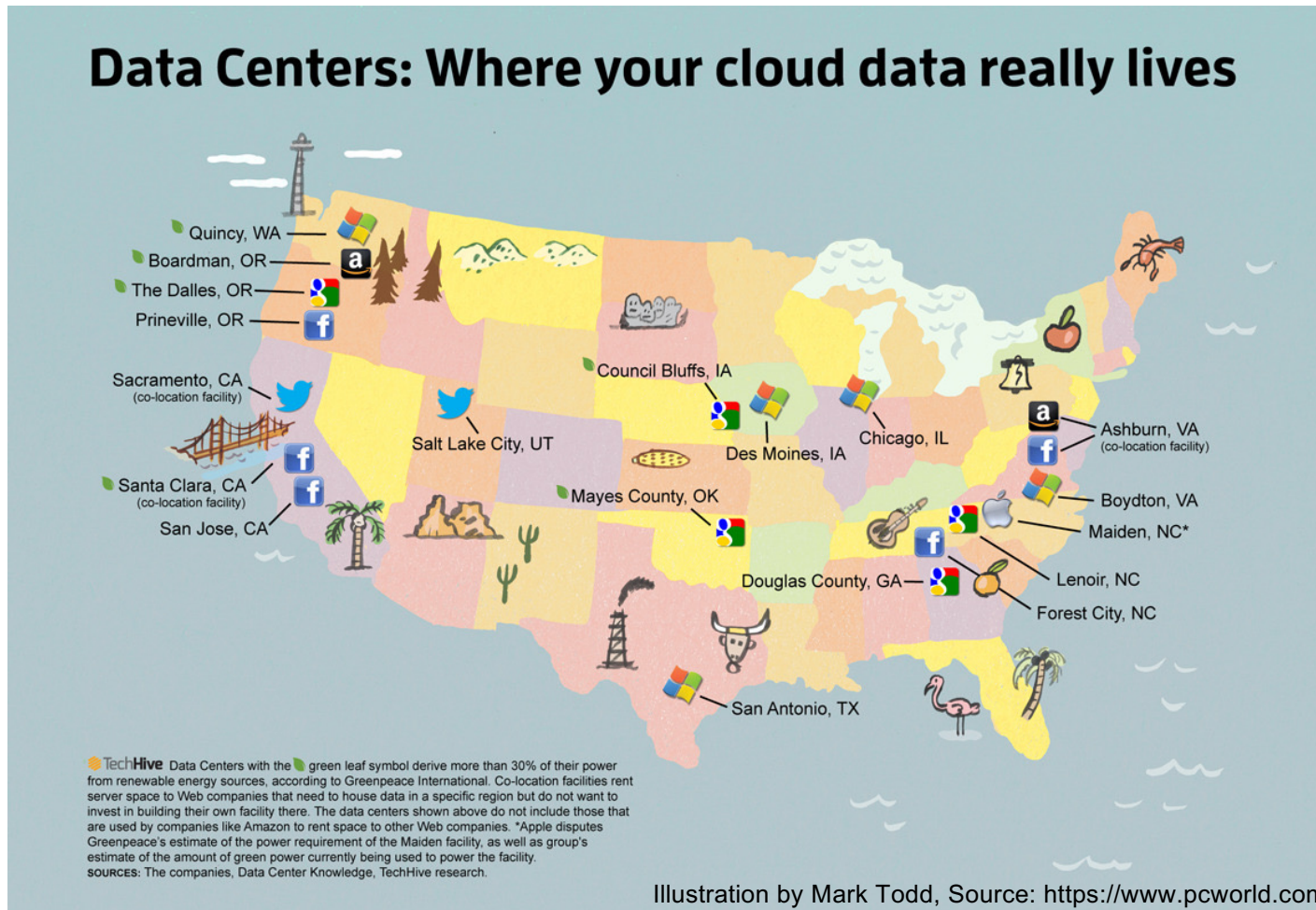


Source: MIRflickr

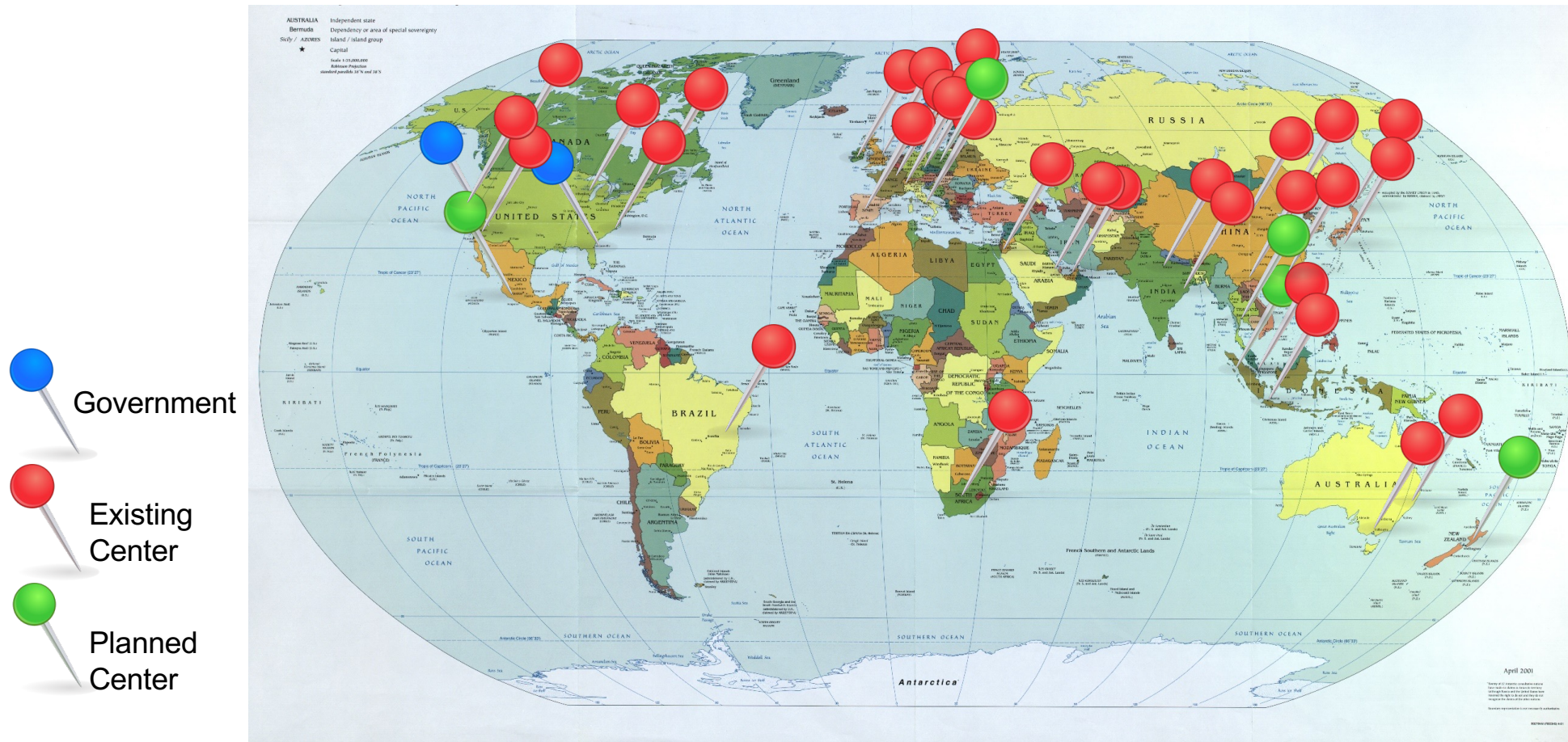
Big Data Applications (Examples)

- Physics
 - During LHC Run 2, the CERN Data Centre processed on average one petabyte (one million gigabytes) of data per day, after initial data reduction by the experiments.
 - This will grow to ~600 PB per year in the next series of experiments (source: [home.cern](#))
- Gaming
 - Fortnite's Live Game Analytics processes 92 million events/min = 2 petabytes/month ([zdnet.com](#))
- Logistics
 - FedEx processes 2 petabytes of new data per month (source: [zdnet.com](#))
- Healthcare
 - A hospital generates ~50 petabytes of data/year (patient records, imaging, operational data, etc.)
 - 97% of this data goes unused, but is nevertheless stored (source: [statnews.com](#))
- Banking
 - Global banks like JPMorgan Chase, Bank of America, and HSBC process 5 - 10 petabytes of transactional, customer, and market data each day (source: [harmonyhit.com](#))

Distributed Systems in Practice: the Cloud



Distributed Systems in Practice: the Cloud (Amazon AWS Data Centers)



Source: <https://aws.amazon.com/about-aws/global-infrastructure/>

Global Overview: <https://www.datacentermap.com/>

Amazon Mechanical Turk (<https://www.mturk.com>)





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Fall 2025

Foundations of Distributed Systems – Applications – FC, HS, CT, IW 2-45

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2.5	Mini-Seminar Topics
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Applications from a Privacy Perspective

- What are relevant applications from a privacy perspective?
 - Well ... basically **all applications** from the other perspectives!
- Let's look at two examples:
 - Covid exposure notification
 - Cryptocurrencies
- In each case:
 - What is the desired functionality?
 - What are the privacy risks if we implement carelessly?
 - How can we do it better?

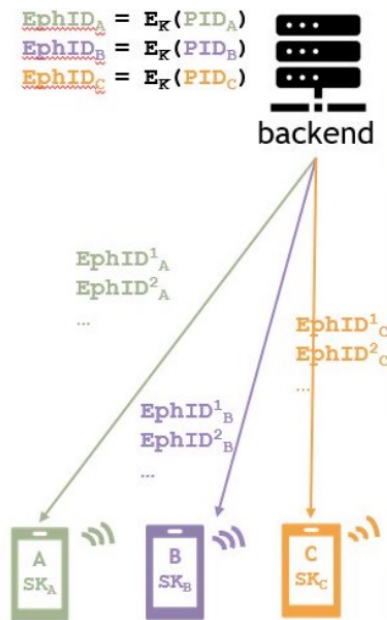
Covid Exposure Notification

- Purpose: alert users who have been in close physical proximity to a confirmed COVID-19 positive case for a prolonged duration
- Exposure event: two users are in close physical proximity for a period of time
- Proximity tracing: process to calculate whom to notify of a high-risk exposure
- Key observation: Covid transmission range shares some similarities with Bluetooth transmission range
 - We can use an [app](#) which automatically records exposure events, performs proximity tracing, and notifies users of exposure

Covid Exposure Notification, the totalitarian way

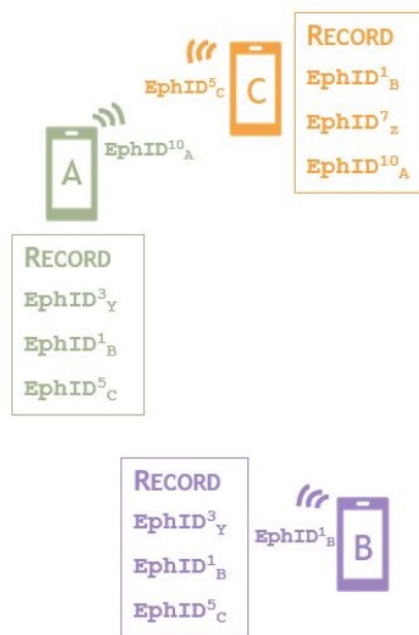
Ephemeral IDs generated centrally, pushed to phones

Installation



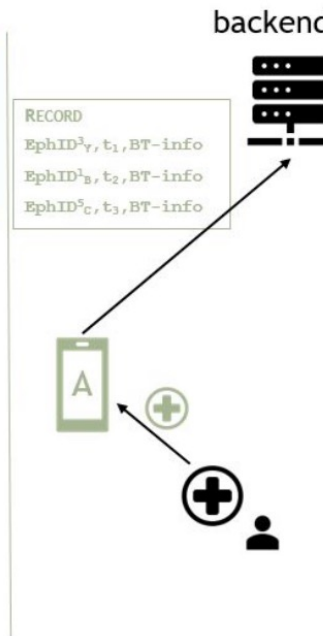
App broadcasts ephemeral IDs, records observed IDs

Normal Operation



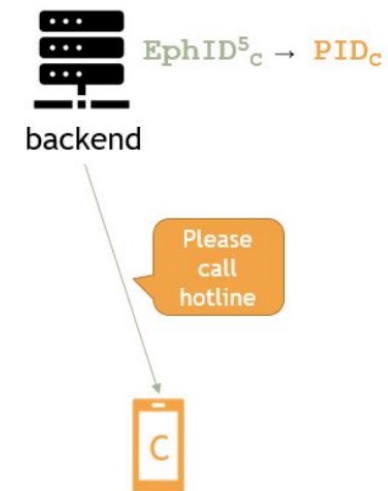
Apps of infected patients upload all observed IDs + metadata

Infection



Server computes risk, pushes notifications to people at risk

Notification

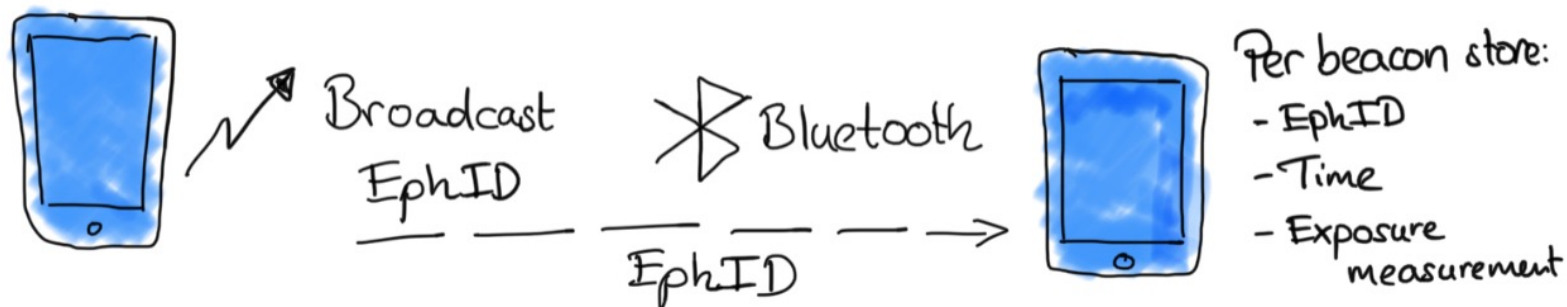


- Server can make “ephemeral” IDs permanent => trace all users over time (e.g., Bluetooth sensors in public places)
- Server can give distinguishable IDs to users => deanonymization and tracking by third parties

- Server learns the contact graph of infected individuals
- Server learns social graphs of non-infected individuals because they occur in the graphs of infected individuals

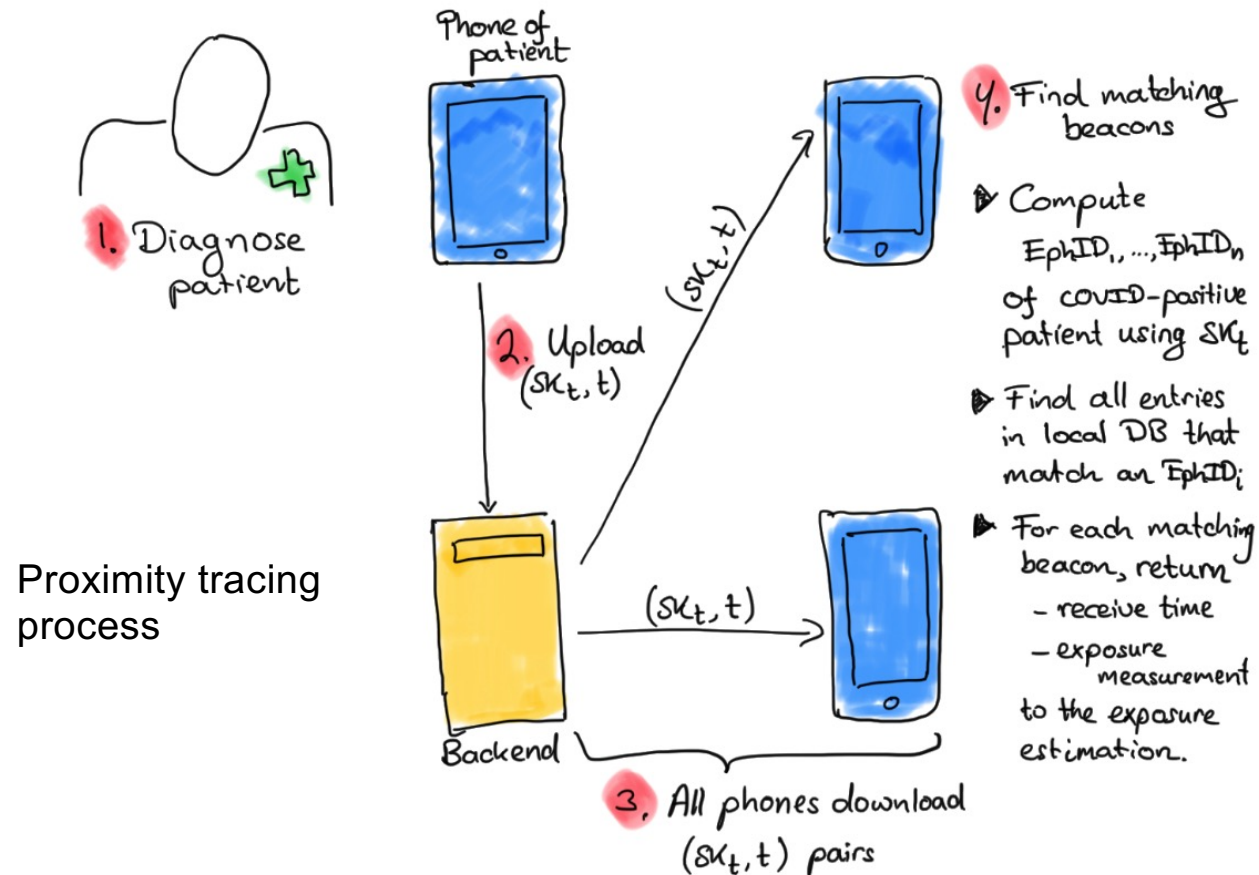
DP3T: Decentralized Privacy-Preserving Proximity Tracing

- Local computation of ephemeral IDs:
 - Each user rotates their secret seed daily: $SK_t = H(SK_{t-1})$
 - $EphID_1 || \dots || EphID_n = PRG(PRF(SK_t, \text{"broadcast key"}))$
 - H : cryptographic hash function
 - PRG : pseudo-random generator (e.g., AES in counter mode)
 - PRF : pseudo-random function (e.g., HMAC-SHA256)
 - Each EphID is broadcast for L minutes (e.g., 15)



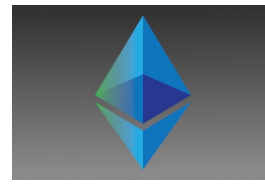
Source: <https://github.com/DP-3T/documents/blob/master/DP3T%20White%20Paper.pdf>

DP3T: Decentralized Privacy-Preserving Proximity Tracing



Source: <https://github.com/DP-3T/documents/blob/master/DP3T%20White%20Paper.pdf>

Cryptocurrencies



- Digital currencies without central authority (unlike regular currencies!)
 - Transactions are stored in a distributed ledger (blockchain)
 - New blocks of transactions are “mined” (finalized) by completing a proof-of-work
- Four “tiers” of privacy in cryptocurrencies:
 1. Pseudonymity: use pseudonyms instead of real names
 2. Set anonymity: user identity can be any from a set of n
 3. Full anonymity: user identity can be any node in the system
 4. Confidential transactions: transacted amounts are hidden

Bitcoin: Pseudonymity



- Users use addresses instead of names:

$$\epsilon = \text{RIPEMD-160}(\text{SHA-256}(\text{publicKey}))$$

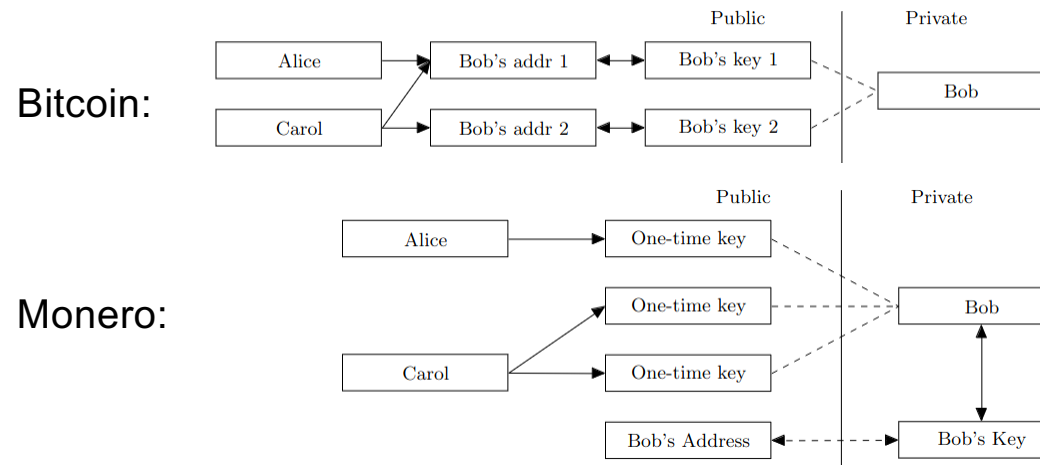
$$\text{address} = \text{version} || \epsilon || \text{checksum}$$

- Additional precautions are possible:
 - New key pair for each transaction
 - Each transaction fully empties input account, creates one or more output accounts (including for change)
- **Clustering attacks** on Bitcoin pseudonymity:
 - Inputs in one transaction are likely owned by the same user
 - Newer output address are likely to be a change address that belongs to the user generating the transaction
 - Map clustered addresses to real-world identities (e.g., online stores)
- **P2P Network attacks** on Bitcoin pseudonymity:
 - Analyze relay patterns of transactions to map Bitcoin addresses to IP addresses

Monero: Set anonymity, confidential transactions



- Unlinkable addresses



- Untraceable signatures

- **Ring signatures:** cryptographic primitive that allows anyone in a group to create a valid signature, without revealing which group member created the signature
- **Ring confidential transactions:** encrypt transaction amounts

- Protect addresses on the P2P network

- Use **Dandelion++**, a protocol with formal anonymity guarantees

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2.5	Mini-Seminar Topics
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Mini-Seminar Organization

- Topics of the mini-seminar: *[see next slides]*
- Form of the seminar presentations
 - Students form groups of 4 (ideally, the same groups as for the exercises)
 - Topic selection implemented via Google Sheets – link will be circulated by email on October 13, 2025 (i.e., in week 5), the topic selection will be due on October 17, 2025.
 - Each group selects one topic and prepares a ~10 min presentation (plus ~5 min Q&A).
- Schedule:
 - Presentations and discussions will take place in weeks #13 and #14
 - Detailed schedule: *tba*
- Note: the mini-seminar presentation and the active participation in the discussions are prerequisites for admission to the oral examination

Topics for FDS Mini Seminar (1/2)

- Idea: present challenges on the following topics from the point of view of all course foci (communication / data / processing / privacy)
- Topics:
 1. Autonomous Vehicles
 2. Blockchain / Distributed Ledgers
 3. CaaS (computing as a service): deplatforming and the smartphone police
 4. Federated and Decentralized Learning
 5. Fediverse
 6. Generative (Visual) Models
 7. LLMs
 8. Local-first vs. M365
 9. Machine Learning Pipelines
 10. Metaverse
 11. Compute Continuum
 12. Smart Governance / open data

Topics for FDS Mini Seminar (2/2)

- Topics (continued):
 13. Energy Footprint of Distributed Systems
 14. Decentralized Energy Grids
 15. Personalized Recommendations
 16. IoT-Based Precision Farming
 17. Cloud-Based Multiplayer Gaming
 18. Digital Twin for Smart Factories
 19. Distributed Live Event Streaming
 20. Autonomous Drone Delivery Networks
 21. Cybersecurity: Distributed Threat Intelligence Sharing
 22. Urban Mobility: Bike and Scooter Sharing Systems
 23. Digital Identity and E-Voting Systems
 24. Real-Time Athlete Performance Tracking in Sports
 25. Distributed Climate Modeling
 26. Digital Cultural Heritage

Topic Choice and Report

- Topic distribution will be done via Google Sheets.
 - The link will be sent on October 13, 2025
 - Entries are due Friday, October 17, 2025
- For each topic, there will be a mentor (will be assigned after topic selection)
Details on the seminar presentation shall be discussed with the mentor
- Report: As a side product of your seminar preparation, a short report (max. 3 – 5 pages) shall be written and handed in by December 19, 2025 (so that you can incorporate details from Q&A) via ADAM.
Typically, having the report first makes slide writing easy. Please aim at a self-contained exposé having ordinary text paragraphs, instead of a list of bullet points.